

Private Fund Holdings

Data-State Analysis

Objective 1 — completeness and cleanliness scorecard across field, fund and manager
DSI Capstone · Phase 1 deliverable

52,048

HOLDINGS

385

FUNDS

101

MANAGERS

8,633

COMPANIES

329

FIELDS

What we were asked to deliver

The Investment Office receives 400+ fund reports as non-standard PDFs. A third-party vendor flattens them into one CSV. Before that CSV can support any analysis, its trustworthiness has to be measured.

MANDATE (PROJECT BRIEF)

- Identify missing data and its cause
- Catch the same company reported under different names
- Flag fields that should always be populated but are not
- Design repeatable procedures Operations can run

THIS DELIVERABLE (OBJECTIVE 1)

- Completeness scored by field, by fund, by manager
- Gap concentration quantified and ranked
- Flagged-record extract for GP follow-up
- Entity-resolution scope sized for Phase 2

Success test from the brief: every major field assessed, and findings clear enough for Operations to prioritise remediation without a data scientist in the room.

Method

01

Load and inspect

837 columns profiled; zero-information attributes dropped

02

Exclude non-investments

Cash, other assets and liability lines removed from scoring

03

Score by field

Fill rates overall and by functional category

04

Score by fund

385 vehicles ranked; outliers isolated at $\pm 2\sigma$

05

Score by manager

101 GPs ranked; within-GP fund variance measured

06

Size entity resolution

Companies appearing under more than one fund counted

07

Export

Flagged records and scorecards written for Operations

SCORING RULE

Completeness of a fund =

non-null mandatory cells

real rows $\times$ 7 fields

Seven mandatory fields:

Deal Status • Current Cost •

Unrealized Value • Capital Invested

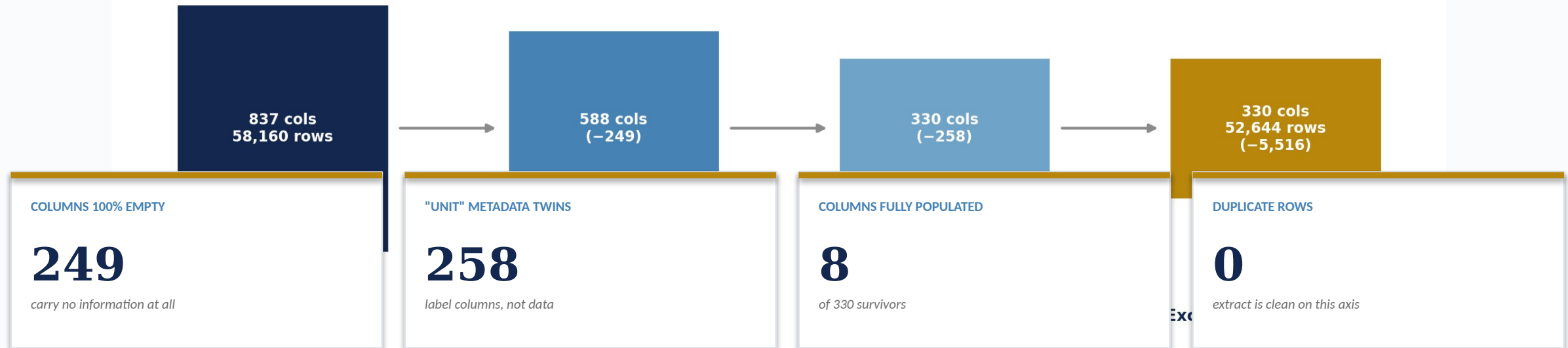
Sector • Geographic Focus • Total Value

GUARDRAIL

Managers with fewer than 20 rows are reported separately. Three holdings and one blank Sector should not read as a reporting failure.

From 837 columns to 330 analysable fields

Cleaning pipeline: 837 columns reduced to 330 analysable fields



Two-thirds of the schema is noise. The vendor emits a fixed superset of columns for every fund regardless of strategy, so sparsity here is a schema artefact, not a GP failure — which is why fill rates must be computed after this reduction, not before.

One column, three different kinds of thing

The vendor puts every line of a GP schedule into Source Asset. Scoring and entity resolution need different subsets of it.

Class	Example	Rows	Score it	Resolve it
holding	Stripe · Valar Co-Invest 13 LP (Qonto)	51,770	Yes	Yes
unnamed_aggregate	Other Investments · Seed Investments	278	Yes	No
balance_sheet	Non-Investment Assets — Cash	5,518	No	No
accounting_entry	Quarterly Unrealized Gain/Loss	577	No	No
subtotal	Subtotal Private Investments	17	No	No

WHAT THE OLD FILTER MISSED

It matched only the Non-Investment prefix. 594 rows of journal entries arriving without that prefix stayed in the population — including a singular "Non-Investment Asset - Cash" the regex did not cover.

WHY AGGREGATES ARE SPLIT OUT

"Other Investments" is real exposure and belongs in a score, but it is not an entity. Counting it as a company puts work in the resolution queue that can never be completed.

Matching is anchored to the whole string, not substrings: the book contains real companies called Itz Cash, Keeper Tax Inc. and Gamma Labs (dba Column Tax). SPV and co-invest vehicles are kept as holdings.

Headline results

PORTFOLIO COMPLETENESS

90.6%

mean across 385 funds

FUNDS AT 100%

118

30.6% of the book

FUNDS BELOW -2σ

19

none above $+2\sigma$

MANAGERS BELOW 85%

14

of 87 scored

CROSS-FUND COMPANIES

1,107

entity-resolution scope

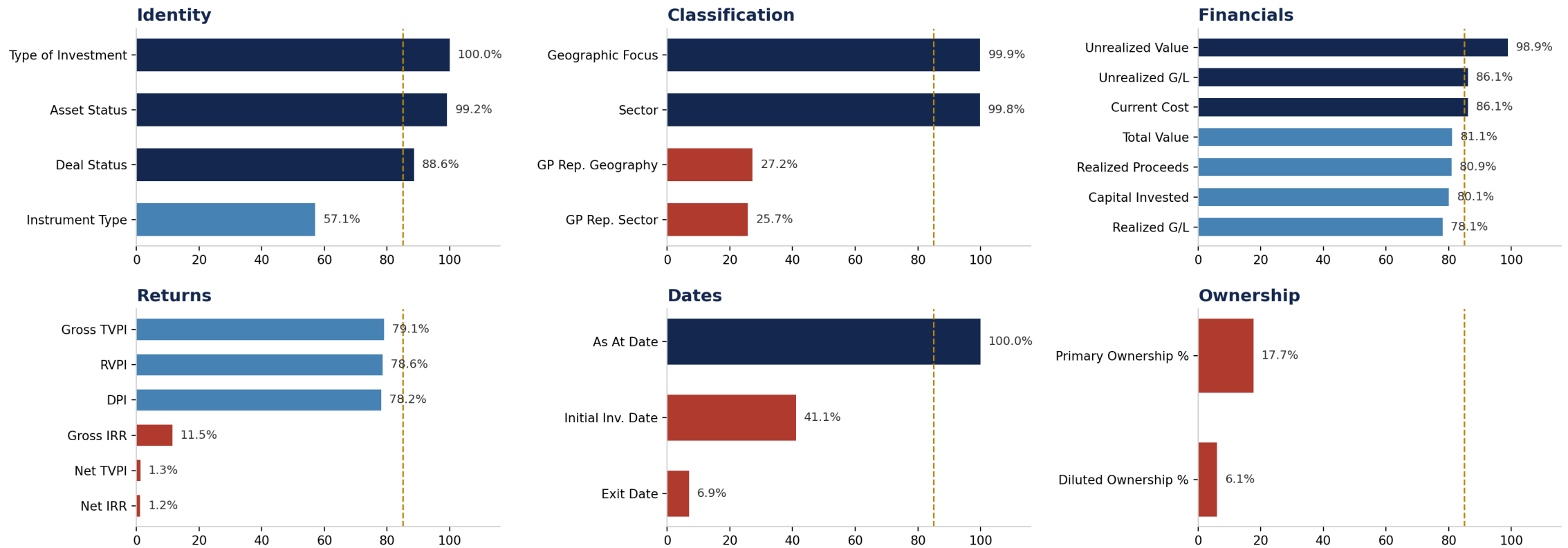
FLAGGED RECORDS

13,176

ready for GP follow-up

One number hides the story. 90.6% average completeness looks healthy, but the distribution is one-tailed and the variance sits inside managers, not between them — the next eleven slides take that apart.

Completeness is not uniform across field families



Identity and classification fields are effectively complete. Financial fields cluster in the low-80s. Returns and ownership fields collapse — and that collapse is the single largest analytical constraint in the dataset.

The returns cliff

Gross-level return metrics are broadly reported. Net-level metrics are effectively absent.

Return metric	Fill rate	Usable?
Gross TVPI	79.1%	Yes
RVPI	78.6%	Yes
DPI	78.2%	Yes
Gross IRR	11.5%	No
Net TVPI	1.3%	No
Net IRR	1.2%	No

WHY THE GAP IS STRUCTURAL

Net returns are fund-level figures net of fees, carry and expenses. They belong on the capital account statement, not the schedule of investments the vendor parses.

This is an extraction-scope gap, not a GP reporting failure.

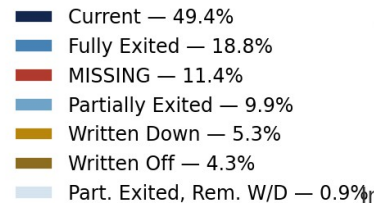
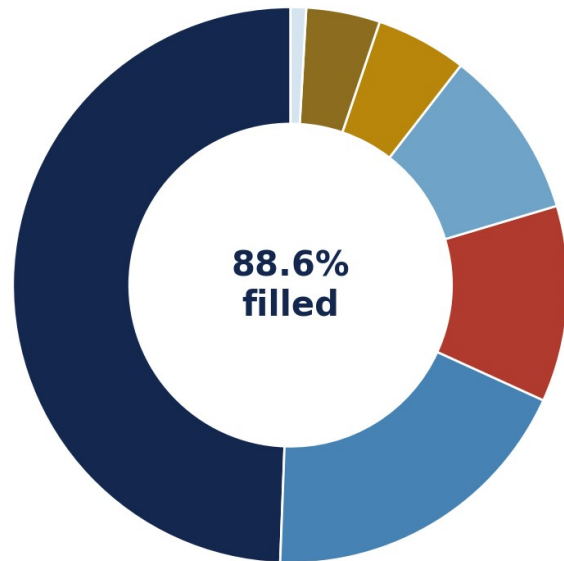
CONSEQUENCE FOR THE INVESTMENT OFFICE

Any net-of-fee performance analysis cannot be sourced from this extract at holding level. Either the capital account reports enter the extraction scope in Phase 2, or net performance stays a fund-accounting output and the holdings CSV is used strictly for gross, position-level work. This should be an explicit decision, not a discovery made mid-analysis.

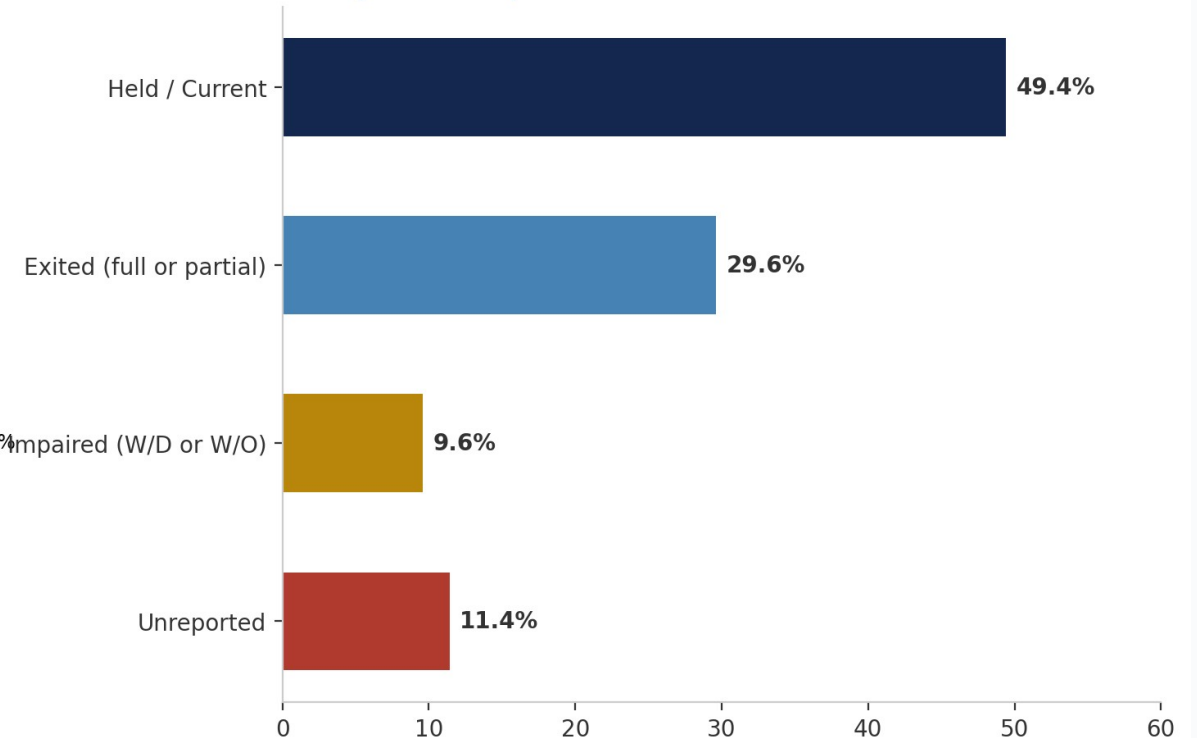
Recommendation: confirm with the vendor whether capital account statements are in extraction scope before any net-return workstream is committed.

Deal Status — the field the brief singles out

Deal Status mix (52,644 rows)



Lifecycle roll-up



The brief flags Deal Status as a field that should always carry a value. It is 88.6% filled: 6,639 rows carry no lifecycle marker at all. Those rows cannot be routed to either the current-portfolio view or the realisations view, so they drop silently out of both.

Systematic gaps versus scattered gaps

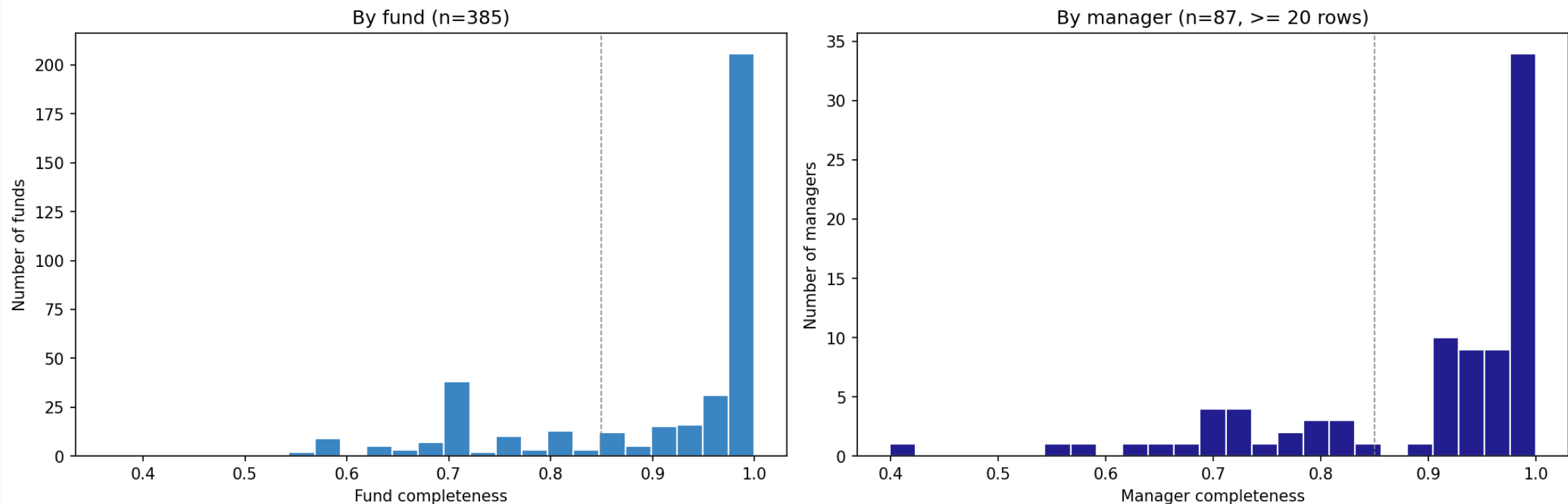
A field 90% complete everywhere is a data problem. A field 100% complete in some funds and 0% in others is a policy choice. Counting funds at zero separates the two.

Mandatory field	Funds at 0% fill	Reading
Sector	0	Universal — no fund omits it
Geographic Focus	0	Universal
Unrealized Value	0	Universal
Deal Status	3	Near-universal; 3 funds opt out entirely
Current Cost	17	Small opt-out cohort
Total Value	58	Structural — 15% of funds never report it
Capital Invested	67	Structural — 17% of funds never report it

Capital Invested and Total Value are not degraded fields — they are absent-by-design for a sixth of the book. Chasing individual blank cells here would waste Operations time; the question is which fund types systematically exclude them and whether that is contractually expected.

Fund and manager completeness distributions

Completeness distribution by dimension



Both distributions are left-skewed against a hard ceiling at 100%. The mass sits above the 85% line, which means remediation is a tail problem — a small number of named vehicles, not a portfolio-wide programme.

Where the funds sit

Completeness band	Funds	Share	Action
100%	118	30.6%	None
95-99%	92	23.9%	None
85-94%	74	19.2%	Monitor
75-84%	63	16.4%	Review
50-74%	35	9.1%	Escalate
Below 50%	3	0.8%	Escalate

THE THREE WORST VEHICLES

Ab3ea87 — 36.5% ($z = -4.05$)

Ae7bb35 — 42.3% ($z = -3.62$)

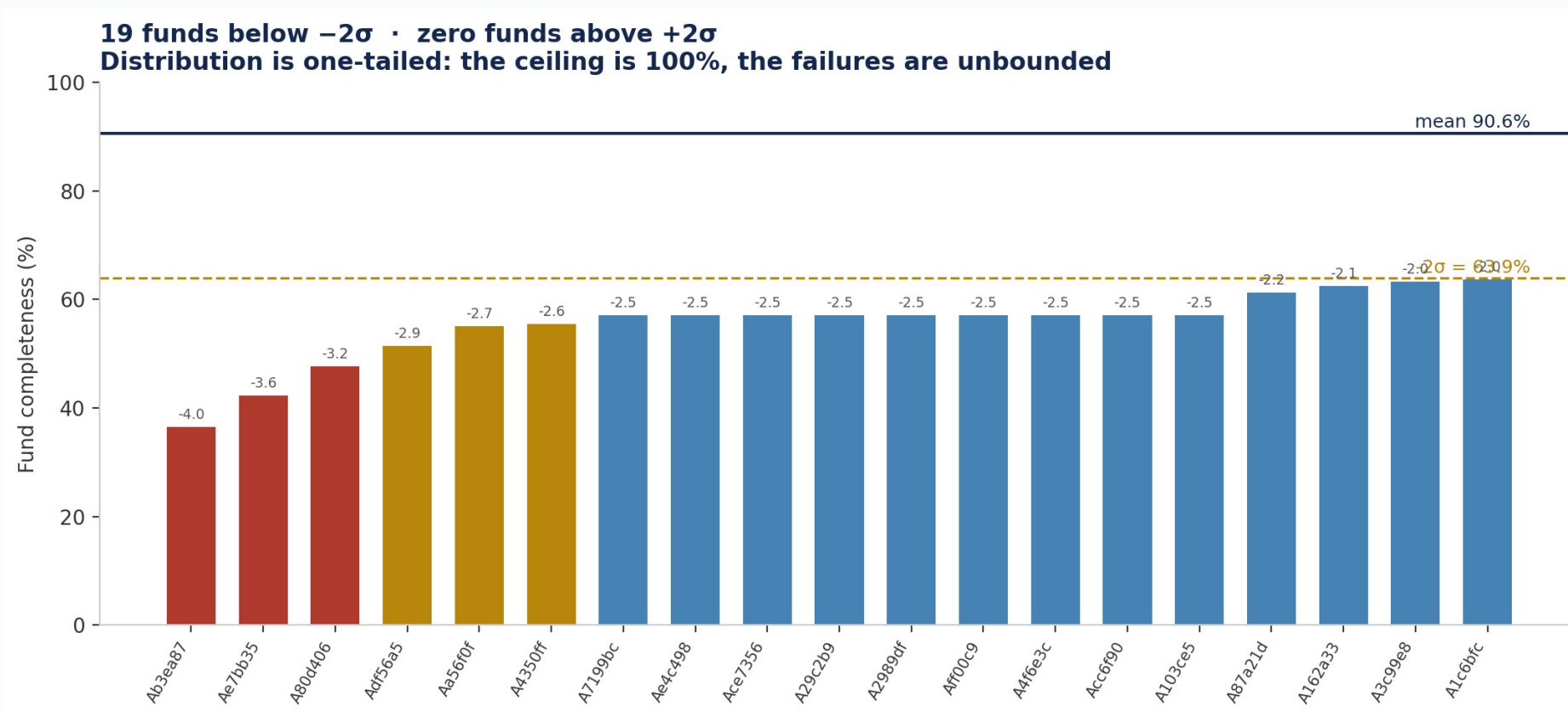
A80d406 — 47.6% ($z = -3.22$)

Each sits more than three standard deviations below the mean.

73% of funds already clear the 85% bar. The remediation target is the 38 funds below 75% — a list short enough to work through by name in a single quarter.

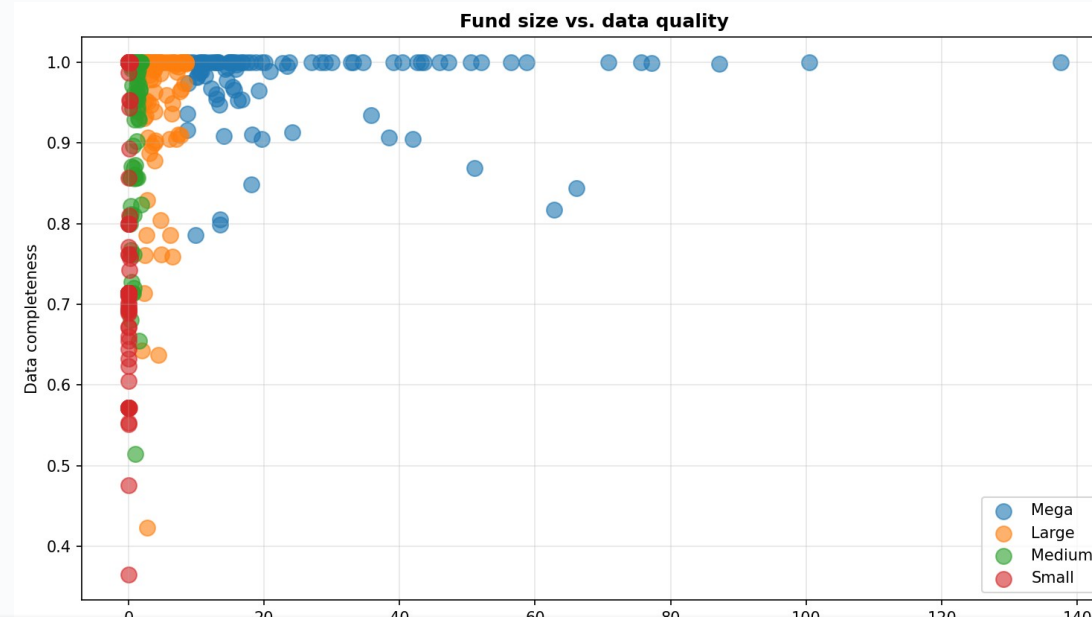
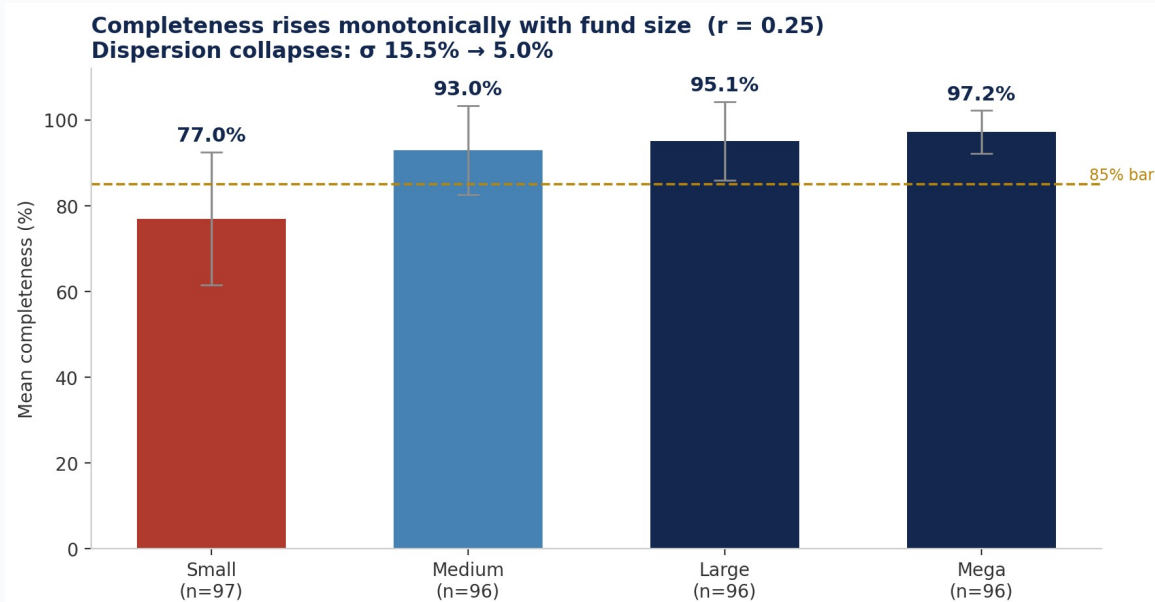
Framing for Operations: this is not a data-quality crisis. It is a concentrated tail of 38 named vehicles, of which 3 are severe.

The distribution is one-tailed



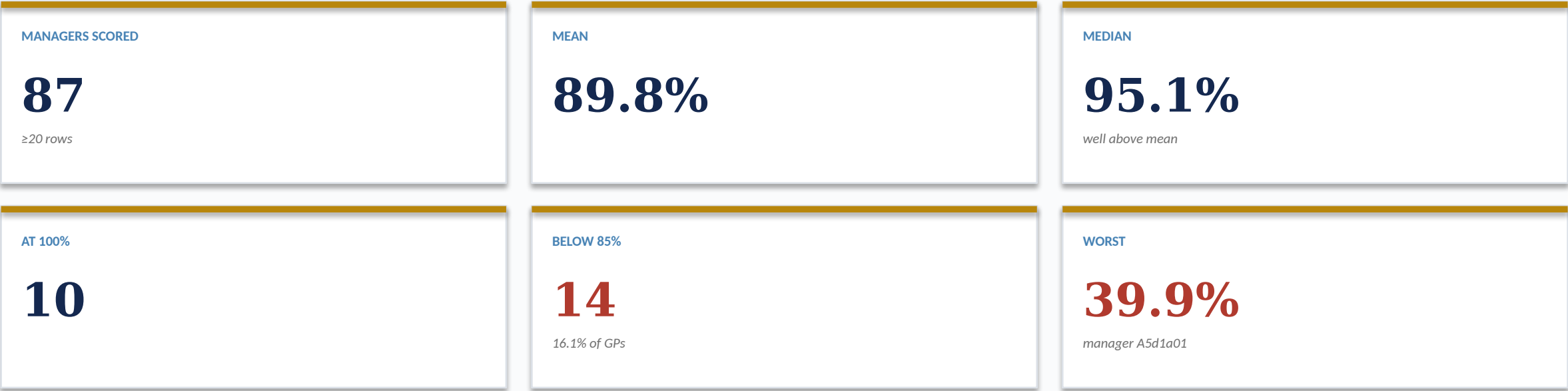
Nineteen funds fall below minus two sigma; none rise above plus two. That asymmetry is mechanical — completeness is capped at 100%, so excellence cannot be an outlier while failure is unbounded. Nine funds cluster at exactly 57.1%, which is 4 of 7 mandatory fields: the same four, suggesting a shared template rather than nine independent lapses.

Larger funds report better — but size is not the mechanism



Mean completeness climbs monotonically from 77.0% to 97.2% across size quartiles, and dispersion collapses from σ 15.5% to 5.0%. But correlation is only 0.25 and small funds span the full range — several report at 100%. Size proxies for back-office maturity; it does not determine outcomes, so it cannot be used to excuse a small manager.

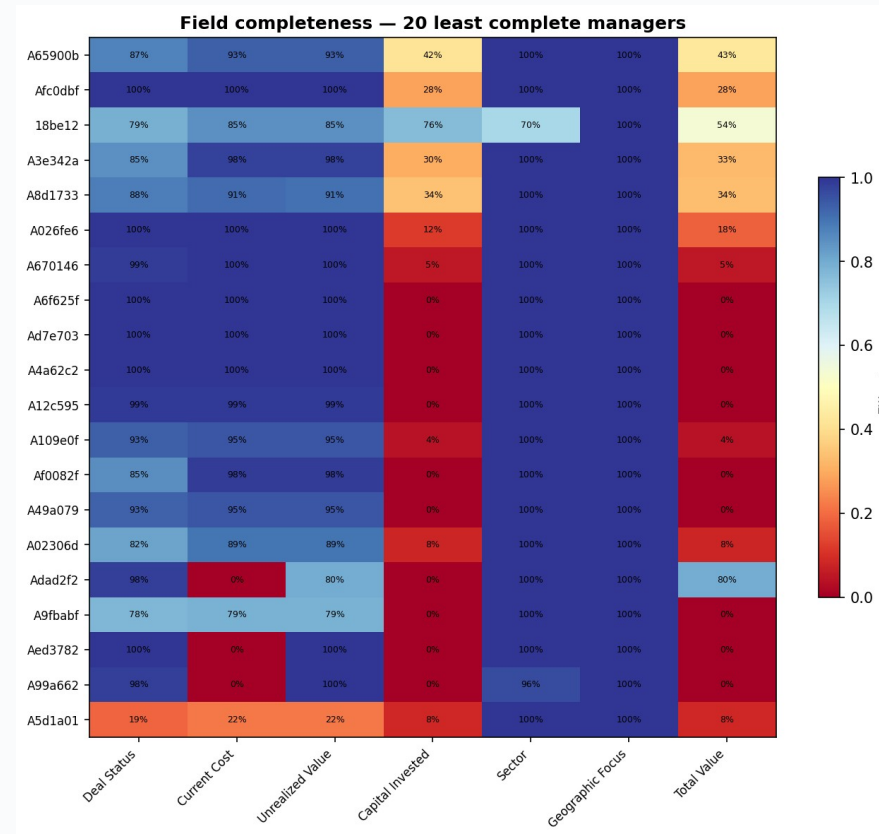
Manager scorecard



Least complete managers	Rows	Funds	Completeness	Worst field
A5d1a01	169	2	39.9%	Deal Status 18.9%
A99a662	54	3	56.3%	Current Cost 0%
Aed3782	226	1	57.1%	Current Cost 0%
A9fbabf	173	3	62.3%	Capital Invested 0%

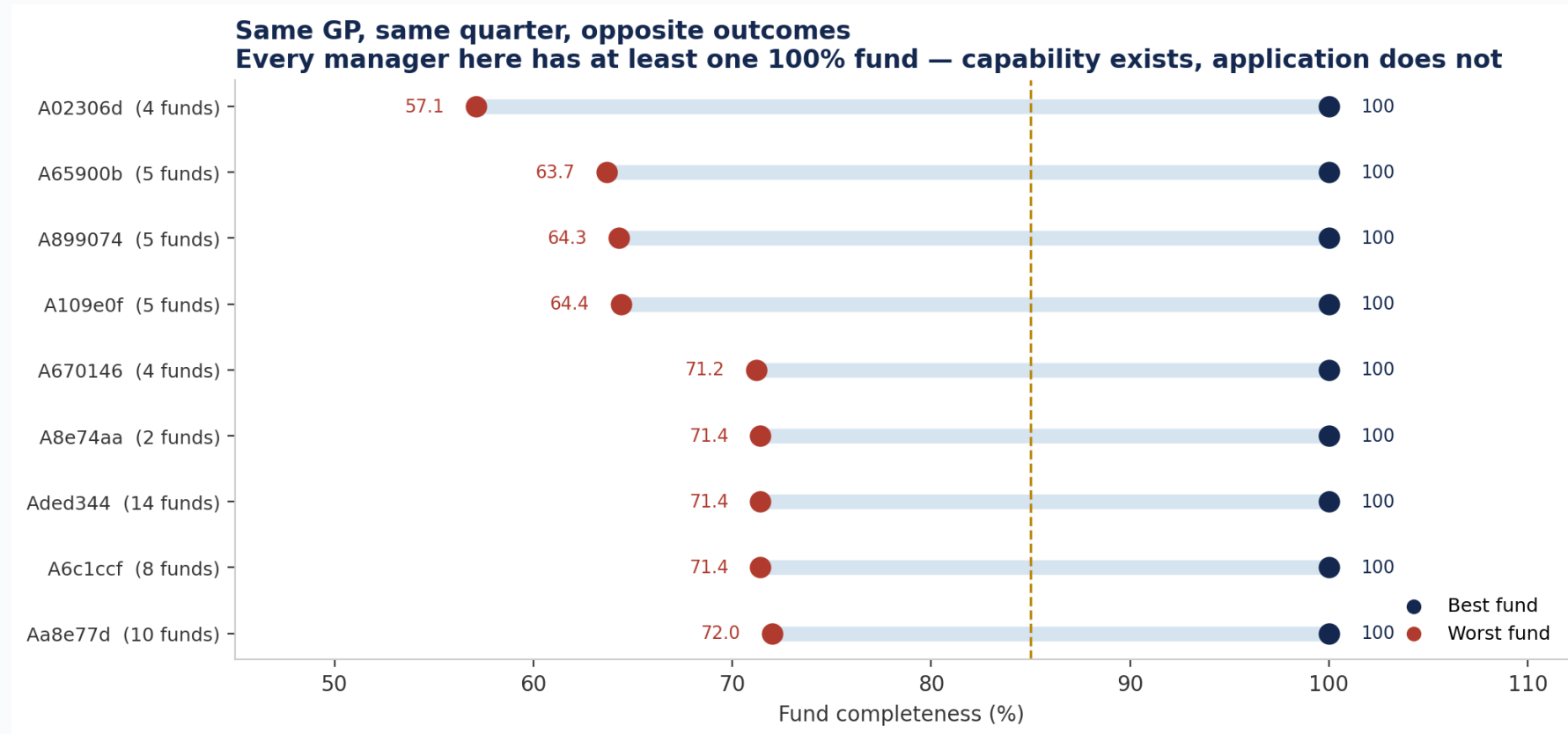
Median far above mean confirms a small tail dragging the average. Seven of the bottom ten share the same worst field: Capital Invested.

Which field does each weak manager omit?



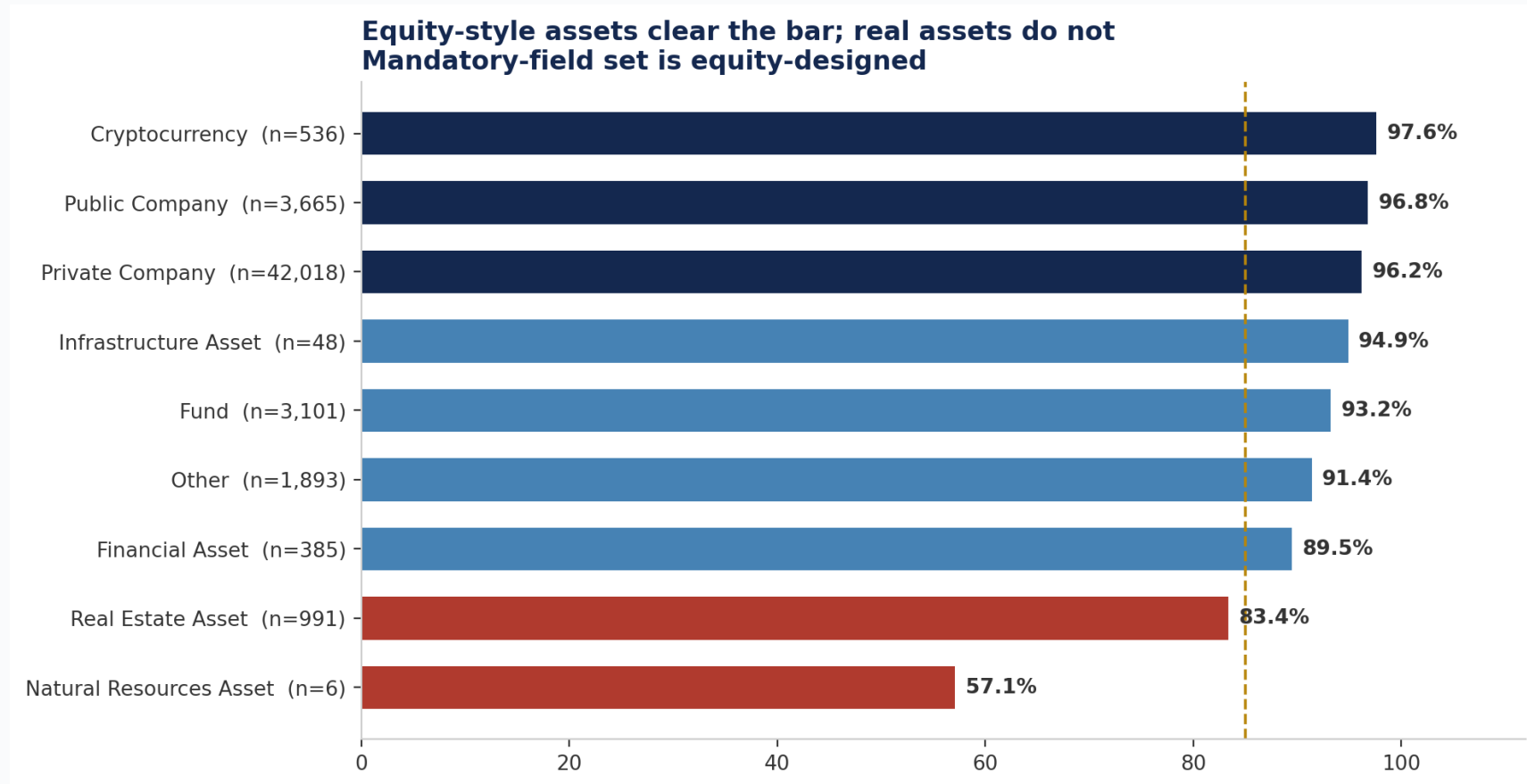
The gaps are vertical, not diffuse. Sector and Geographic Focus stay green across every weak manager; Capital Invested is red almost everywhere. A manager at 60% overall is not sloppy across the board — they are missing one or two specific fields, which makes the follow-up request a single sentence rather than a data-quality lecture.

The most actionable cut in the analysis



Every manager shown has at least one fund at 100% and another below 72%. Same GP, same reporting quarter, opposite results. The capability plainly exists, so these are fund-level breakdowns to raise about a named vehicle — not manager-level policy conversations. Managers with near-zero spread are the opposite case and need the policy conversation instead.

Quality by investment type



Equity-style holdings clear the bar comfortably. Real Estate at 83.4% and Natural Resources at 57.1% sit below it — and both are being measured against a mandatory field set built for equity.

Real assets fail an equity-shaped test

Mandatory field	Equity meaning	Real-asset reality
Deal Status	Current / Exited / Written off	A property is held, leased or listed for sale — no exit event
Current Cost	Entry cost of the position	Book value plus capitalised improvements
Unrealized Value	Mark versus cost	Third-party appraisal on its own cycle
Capital Invested	Staged rounds	One purchase, then ongoing capex
Sector	GICS-style classification	Office / retail / industrial taxonomy

NATURAL RESOURCES — 6 ROWS

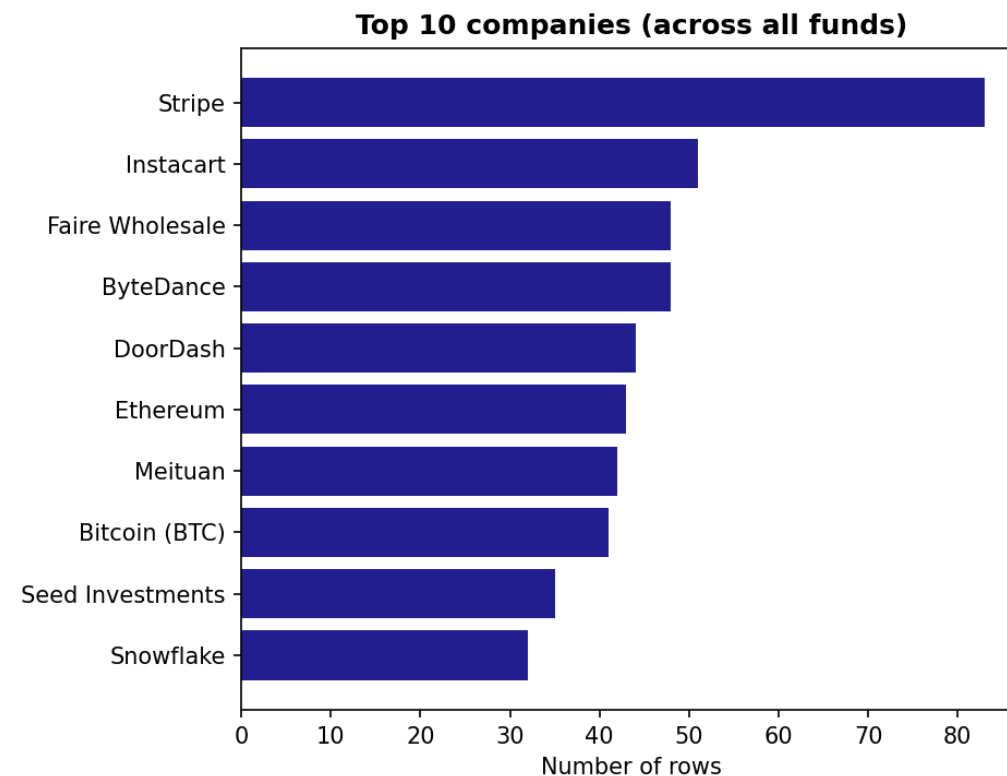
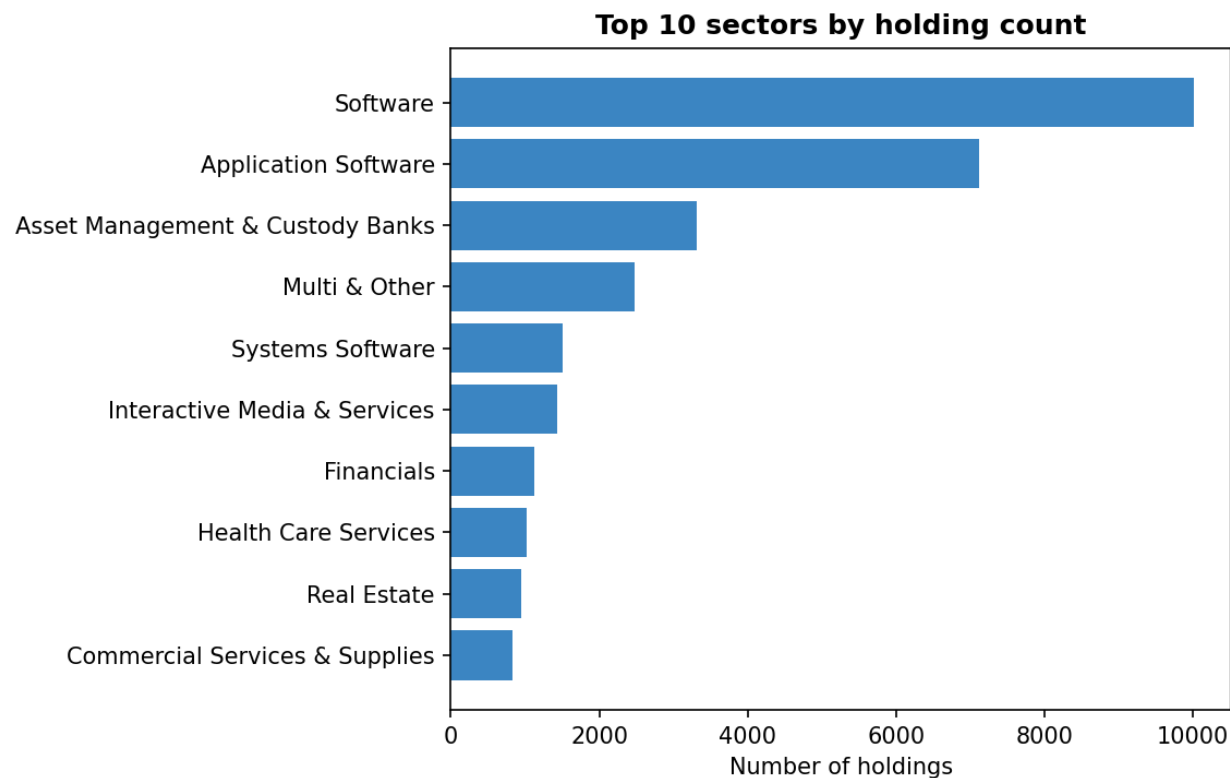
Lifecycle is exploration → development → production → decline. None of it maps to exit or write-down. Sample is too small to score meaningfully in any case.

WHAT THIS IS NOT

This is not evidence that real-asset GPs report poorly. It is evidence that the yardstick was built for a different asset class.

Recommendation: define an asset-class-specific mandatory set before these managers appear on any remediation list. Scoring them against the current set produces a false positive.

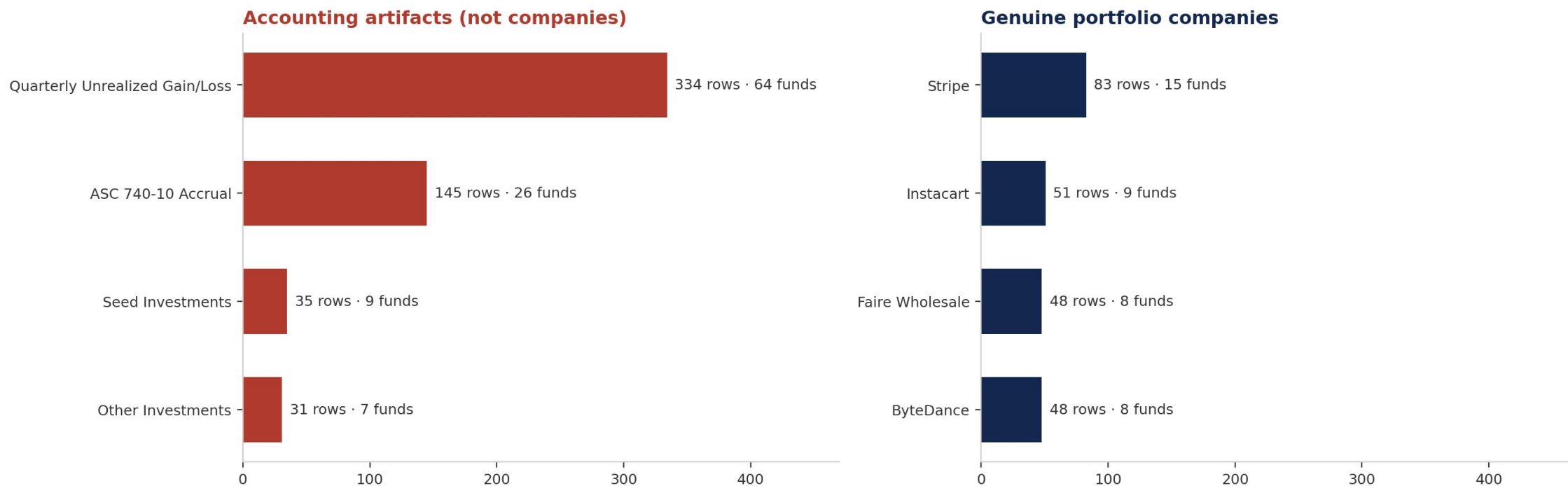
Portfolio concentration



Software and Application Software alone account for 32.5% of holdings; the top five sectors reach 44.6%. Concentration this high means a single sector-taxonomy error propagates across a third of the book.

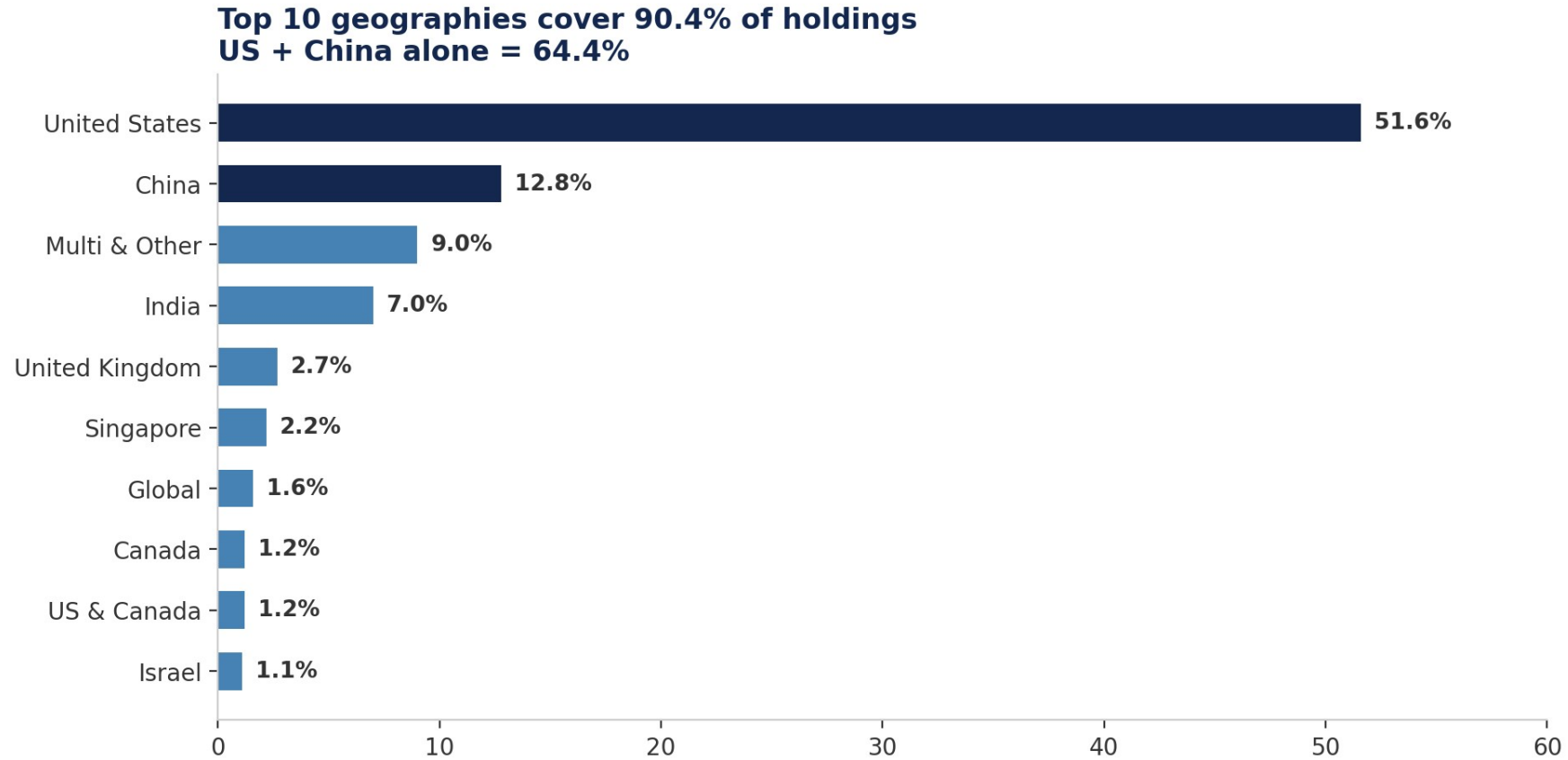
The two most-reported "companies" were journal entries

The two most-reported "companies" in the book are journal entries



"Quarterly Unrealized Gain/Loss" appeared 334 times across 64 funds and "ASC 740-10 Accrual" 145 times across 26, both sitting in Source Asset as though they were portfolio companies. The classifier now removes 594 such rows, and both entries drop out of the entity resolution queue where they had been the two largest items.

Geographic distribution



Reported completeness reaches 100% in Africa, Peru, Estonia and Nigeria — but each rests on a handful of rows. Small-n geographies should be read as noise, not as reporting excellence.

Entity resolution — sizing the Phase 2 problem

RESOLVABLE NAMES

8,633

after classification

HELD BY MORE THAN ONE FUND

1,107

12.8% of names

MEAN APPEARANCES PER NAME

6.0

across quarters and funds

WHY THE COUNT OVERSTATES THE COMPANY UNIVERSE

Two funds holding the same business can write it two ways. "Oomnitz" on one schedule of realised gains and "Oomnitza, Inc." on another schedule of investments are the same company — the extract treats them as two.

The 1,107 multi-fund names are the working set: these are the only ones where a naming mismatch can split a single company into several, and the only ones where cross-fund valuation comparison is possible at all. Twenty-seven names were removed from this queue by the classifier, led by the two journal entries on the previous slide.

Phase 2 dependency: deal-status inference cross-references the schedule of investments against the schedule of realised gains. That join is by company name, so canonical mapping has to land before inference can run. Fuzzy matching may propose candidates; it may not decide them.

Findings ranked

#	Finding	Scale	Severity	Owner
1	Capital Invested absent by design	67 funds (17.4%)	High	Vendor + GP
2	Net-return metrics unusable	Net IRR 1.2%	High	Vendor scope
3	Managers below 85%	14 of 87	High	Operations
4	Entity resolution unresolved	1,107 names	Medium	Phase 2
5	Accounting entries as companies	594 rows	Resolved	Fixed
6	Deal Status blank	6,639 rows	Medium	GP follow-up
7	Real assets mis-scored	997 rows	Low	Redefine metric
8	Severe fund outliers	3 funds below 48%	High	Operations

Two of the top three findings are not GP failures — they are extraction-scope and field-definition questions that belong with the vendor. Sequencing those first avoids sending Operations to chase data the GPs already sent.

Recommendations

01	Resolve Capital Invested and Total Value with the vendor Confirm whether the 67 and 58 zero-fill funds are contractual exclusions or extraction misses. Nothing else should be chased until this is settled.	Vendor	Immediate
02	Decide the net-return question explicitly Bring capital account statements into scope, or record that net performance is out of scope for this extract.	Vendor	Immediate
03	Tier manager outreach Tier 1: 3 GPs below 60%. Tier 2: 11 GPs at 60–85%. Tier 3: high-spread GPs, raised per fund rather than per relationship.	Operations	Q1
04	Define an asset-class mandatory set Real Estate and Natural Resources scored against their own field list before either appears on a remediation list.	Analytics	Q1
05	Extend the non-investment filter ✓ DONE Three-way classifier shipped with 57 regression tests. 594 rows reclassified; 27 names left the resolution queue.	Analytics	Complete
06	Stand up canonical naming Begin with the 1,107 multi-fund names. Explicit aliases, reviewed, not auto-accepted.	Analytics	Q1–Q2
07	Re-run quarterly The notebook is parameterised. Track band migration rather than the headline average.	Operations	Ongoing

Phase 1 closes, Phase 2 opens

PHASE 1 — DESCRIPTIVE ✓ COMPLETE

"What state is the data in?"

- Completeness scored on three axes
- Gap concentration quantified
- 13,176 records flagged
- Entity-resolution scope sized
- Eight exports written for Operations

Cannot answer: is the data correct?

PHASE 2 — VALIDATIVE NEXT

"Is the data correct?"

- Parse schedule of investments from source PDFs
- Reconcile cost and fair value to fund totals
- Infer Deal Status across schedules
- Compare vendor CSV against statements
- Attach page and bounding box to every value

Blocked on: canonical entity naming

The two phases answer different questions. Completeness says a value is present; validation says it is right. A fund at 100% completeness carrying wrong numbers still scores 100% here — which is precisely why Phase 2 exists.

Exports and validation

File	Contents
data_state_by_field.csv	Fill rate, all 329 value columns
data_state_by_fund.csv	Completeness, all 385 funds
data_state_by_manager.csv	Completeness, rows, funds — 101 GPs
data_state_by_manager_field.csv	Field-level fill per manager
flagged_missing_fields.csv	13,176 records for GP follow-up
source_asset_classification.csv	Row and name counts per class
excluded_non_investment_rows.csv	6,112 rows removed, with reason

REPRODUCIBILITY

The notebook runs end to end from the raw extract with no manual steps. Thresholds — the 85% bar, the 20-row manager floor, the mandatory field list — are parameters at the top of the file, so a change in Investment Office policy is a one-line edit and a re-run, not a rewrite.

ASSERTIONS PASSED

- Fund completeness within [0,1]
- Manager completeness within [0,1]
- No duplicate rows
- Unit columns stripped
- Flagged $\subseteq$ all rows

Open question for the Investment Office: is 85% the right bar, and should Sector carry equal weight to Current Cost in the mandatory set? Both are currently assumptions.

Questions

Phase 1 data-state analysis complete.

Phase 2 validation blocked only on canonical entity naming.